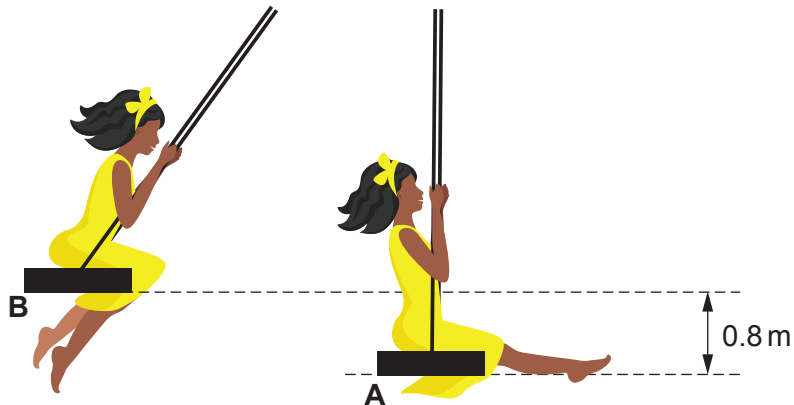


4. A child of mass 35 kg is playing on a swing.



The swing is pulled back to position **B**, 0.8 m above the start point, and released.

- (a) (i) Use an equation from page 2 to calculate the potential energy gained by the child when moving from position **A** to position **B**. [2]
[Gravitational field strength, $g = 10 \text{ N/kg}$]

potential energy = J

- (ii) State the kinetic energy of the child as they pass through point **A**. Assume there are no resistive forces acting. [1]

kinetic energy = J

- (iii) Use an equation from page 2 to calculate the velocity of the child as they pass through point **A**. [3]

velocity = m/s



- (b) (i) At position **B** the child drops a ball which hits the ground time $t = 0.50$ s later. The ball has an initial velocity, $u = 0$ m/s. The downwards acceleration, $a = g = 10$ m/s².

Use an equation from page 2 to determine the distance, x , that the ball falls. [3]

$x = \dots\dots\dots$ m

- (ii) After the collision with the ground, the ball rolls away with initial kinetic energy of 6.4 J and then comes to a halt in 3.0 m.

Use an equation from page 2 to calculate the size of the friction force between the ball and the ground. [2]

friction force = $\dots\dots\dots$ N

